THE UNITED REPUBLIC OF TANZANIA
THE PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT
NAMANGA FORM THREE PRE-MIDTERM EXAMINATION

BASIC MATHEMATICS

MARKING SCHEME AUGUST 2024

SECTION A (60 MARKS)

1. La L(a) i. Factors of 96 are

2	96
2	48
2	24
2	12
2	6
3	3
	1 .....$00\frac{1}{2}$ mark

Direct observed factors of 96 are 1,2,3,6,12,24,48, and 96
Indirect observed factors of 96 are obtained by multiplying two or more of prime factors from the table
$2\times 2 = 4, 2\times 2\times 2 \times = 8, 2\times 2\times 2\times 2 = 16,$
$2\times 2\times 2\times 2\times 2 = 32$
Factors of 96 are 1,2,3,4,6,8,12,16,24,32,48, and 96.....$00\frac{1}{2}$ mark.

Average of factors of 96= $\frac{1+2+3+4+6+8+12+!6+24+32+48+96}{12}$....$00\frac{1}{2}$ mark.
$= \frac{252}{12}$.....$00\frac{1}{2}$ mark.
$= 21$

Therefore 21 is the average sum of all factors of 96.....$00\frac{1}{2}$ mark.

i i. $= \frac{2}{5} + \frac{3}{5}\left[\frac{20+27}{36}\right]$

$= \frac{2}{5} + \frac{3}{5}\left[\frac{47}{36}\right]$.....$00\frac{1}{2}$ mark
$= \frac{2}{5} + \frac{3}{5}\times\frac{47}{36}$
$= \frac{2}{5} + \frac{47}{60}$.....$00\frac{1}{2}$ mark.
$= \frac{24+47}{60}$.....$00\frac{1}{2}$ mark
$= \frac{71}{60} = 1\frac{11}{60}$.....$00\frac{1}{2}$ mark.

(b) The numbers are 35, 45, and 55
$35 = 5\times 7$
$45 = 3\times 3\times 5$
$55 = 5\times 11$.....$00\frac{1}{2}$ mark
The GCF is 5 .....$00\frac{1}{2}$ mark.
The LCM $= 3\times 3\times 5\times 7\times 11 = 3\,465$.....$00\frac{1}{2}$ mark.
The sum $LCM + GCF = 3\,465 + 5 = 3\,470$.....$00\frac{1}{2}$ mark.

2. i. $\frac{9^x+9^x+9^x}{3^x+3^x} = 13\frac{1}{2}$

$\frac{9^x(3)}{3^x(2)} = \frac{27}{2}$ ……$00\frac{1}{2}$ mark

$\left(\frac{9}{3}\right)^x = 24\frac{27}{2} \times \frac{2}{3}$

$(3)^x = 9$ ……$00\frac{1}{2}$ mark

$3^x = 3^2$

$x = 2$……01 mark.

ii. $\frac{1}{b^3} = 8$

$1 = 8b^3$……$00\frac{1}{2}$ mark

$\frac{1}{8} = b^3$……01 mark

$\sqrt[3]{\frac{1}{8}} = \sqrt[3]{b^3}$

$b = \frac{1}{2}$……$00\frac{1}{2}$ mark

(b)$\log_y 200 = \log_3 2^3 + \log_3 5^2$

$= \log_3(8 \times 25)$……01 mark

$= \log_3 200$

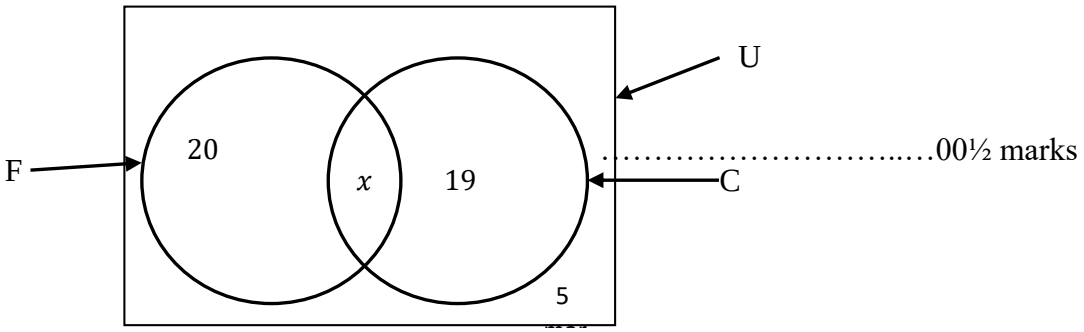
$y = 3$……01 mark

3. (a) Let n(U) = 52

Let the number of students who study French be n(F)

Let the number of students who study Chinese be n(C)

By using s Venn diagram:



……..…00½ marks

Let the number of students who study both languages be x

$20 + x + 19 + 5 = 52$ ……00½ mark

$44 + x = 52 - 44$……00½ mark

$44 - 44 + x = 52.44$

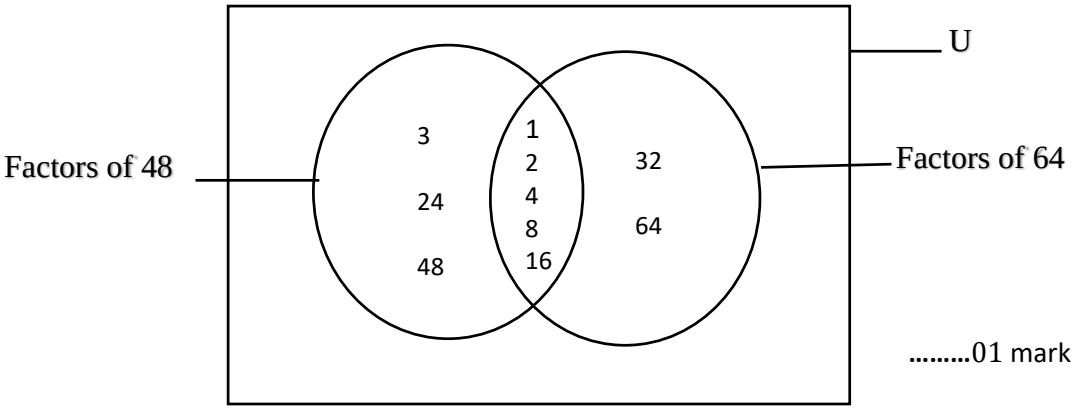
$x = 8$ ……...00½ mark

n(CUF) = 8 + 20 +19 = 47 students ……..00½ mark

(b) i. Factors of 48 are 1, 2, 3, 4, 6, 8, 12, 16, 24, and 48 ……$00\frac{1}{2}$ mark

Factors of 64 are 1, 2, 4, 8, 16, 32, and 64……$00\frac{1}{2}$ mark

Common factors of 48 and 64 are 1, 2, 4, 8 and 16……$00\frac{1}{2}$ mark



……01 mark

ii. 16 is the highest common factor of 48 and624……$00\frac{1}{2}$ mark

4. (a) coordinates are $(-5,0) and\ (0\ ,5)$

$m = \frac{y_2-y_1}{x_2-x_1} = \frac{5-0}{0--5} = \frac{5}{5} = 1$ ……01 mark

$1 = \frac{y-5}{x}$ ……01 mark

$x = y - 5$

$y = x + 5$……01 mark

(b) Let adjacent angle to $2x$ inside the parallelogram be y

$y + 2x = 150$ (Opposite angles in the parallelogram are equal)….(1)…… $00\frac{1}{2}$ mark

$y + x + 90^{\circ} = 180^{\circ}$ (Angle on Straight line) ……$00\frac{1}{2}$ mark.

$y + x = 90^{\circ}$……(2)…… $00\frac{1}{2}$ mark.

Solve equation 1 and 2 by elimination method

$-\left|\begin{matrix} y + 2x = 150 \\ y + x = 90 \end{matrix}\right.$ ……$00\frac{1}{2}$ mark.

$x = 60^{\circ}$……01 mark

5. (a) i. $\Delta PNQ \sim \Delta MRQ$……$00\frac{1}{2}$ mark.

$\frac{PN}{MR} = \frac{NQ}{BQ} = \frac{PQ}{MQ}$ =Constant of proportionality or scale factor

Scale factor $= \frac{11.25}{4.} = 2.8125$……$00\frac{1}{2}$ mark.

$\frac{n}{6} = 2.8125, n = 6 \times 2.8125\ cm = 16.875\ cm$ ……01 mark

$\frac{14.625}{m} = 2.8125, m = \frac{14.625}{2.8125} = 5.2\ cm$……01 mark

(b) $\angle MAP = \angle MAP$ (given)……01 mark.

$\overline{LP} = \overline{MP}$(given)…… $00\frac{1}{2}$ mark.

$\overline{AP}$ is common……$00\frac{1}{2}$ mark.

$\overline{AP}\ bisects \angle LAM$ (By SSS)…… ……01 mark.

6. (a) Let the number of litres of diesel used be l and let d be the distance covered

As the distance covered increases also the number of litres of diesel used increase
Therefore the two quantities are directly related

$d_1 = 60\ km, l_1 = 5\ litres, d_2 = 150\ km. required = l_2$

$d \propto l\ d = lk$

$\frac{d_1}{l_1} = k = \frac{d_2}{l_2}$ ……01 mark

$\frac{60\ km}{5\ l} = 12\frac{km}{l} = \frac{150\ km}{l_2}$ ……. …01 mark

$l_2 = \frac{150\ km\ l}{12\ km}$

$= 12.5\ litres$

Therefore 12. 5 litres are needed to cover a distance of 150 km……..……01 mark.

(b) $v \propto x^2 \times \frac{1}{y}, v = \frac{x^2}{y} \times k, k = \frac{yv}{x^2}$ ……01 mark

$v_1 = 18, x_1 = 3, y_1 = 4, x_2 = 5, y_2 = 2\ , required\ v_2$

$k = \frac{18\times4}{3^2} = 8$ ……01 mark

$8 = \frac{v\times2}{25}\ , v = \frac{8\times25}{2} = 100$…… ……01 mark

7. (a) Buying price $= sh.\ 20\ 000$

Selling price $= sh.\ 29\ 000$

Profit = selling price –buying price

$sh.\ 9\ 000 = sh.\ 29\ 000 - sh.\ 20\ 000$……01 mark.

Percentage profit $= \frac{profid\ made}{buying\ price} \times 100\%$

$= \frac{9\ 000}{20\ 000} \times 100\%$……01 mark

$= 45\%$……01 mar

(b) P$= 1\ 200\ 000$ shillings, Time $= 3\ years$

$R = 4.5 \times 4 = 18\%\ p.a$……01 mark

$I = prt$

$= 1\ 200\ 000 \times \frac{18}{100} \times 3$……01 mark

$= 648\ 000$ shillings……01 mark

8. (a)

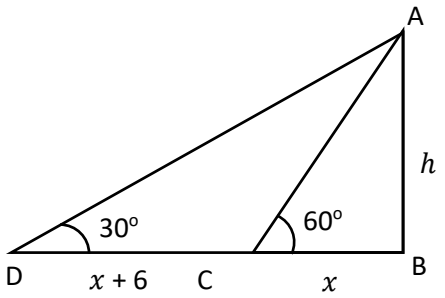
N	Number	Standard form	Logarithm
Numerator	9 804	9.804×10^3	3.9914
	23.19	23.19×10^1	+1.3653
			5.3567
	0.086	8.6×10^{-2}	$\overline{2}.9345$
	41 750	$4..1750\times 10^4$	+ 4. 6207
Denominator			3. 5552
Logarithm of a numerator			5.3567
Logarithm of denominator			- 3.5552

Logarithm of a quotient			1.8015

Use mathematical tables to read the ant-logarithm of 0.1815, then multiply the answer by 10^1 that is 6.331×10^1

Therefore $\frac{9\,804 \times 23.19}{0.086 \times 41\,750} = 63.31$……03 marks

(b)



Let $\overline{bc}$ *be* x, *let* $\overline{DC}$ *be* $(x+6)$

$Tan\ 30^{\circ} = \frac{h}{2x+6}$ ……$00\frac{1}{2}$ mark

$Tan60^{\circ} = \frac{h}{x}$ ……$00\frac{1}{2}$ mark

$(2x+6)tan30^{\circ} = xtan60^{\circ}$

$(2x+6)\frac{\sqrt{3}}{3} = x(\sqrt{3})$……$00\frac{1}{2}$ mark

$3x = 2x + 6$

$3x - 2x = 6$

$x = 6$ ……$00\frac{1}{2}$ mark

$h = x\sqrt{3} = (6\sqrt{3})cm$……01 mark

9. (a) From ΔQNP, $(\overline{QN})^2 + (\overline{PN})^2 = 10^2 + 24^2$

$= 100 + 576 = 676$

$(\overline{QP})^2 = 26^2 = 676$

Since $(\overline{QP})^2 = (\overline{QN})^2 + (\overline{PN})^2 = 676$……01 mark

Then, $\angle QNP = 90^{\circ}$ (Pythagoras theorem)

Hence, $\angle PNR = 90^{\circ}$ (Supplement of, $\angle QNP$) ……01 mark.

Applying Pythagoras theorem to ΔPNR

$(\overline{PN})^2 + (\overline{NR})^2 = (\overline{PR})^2$

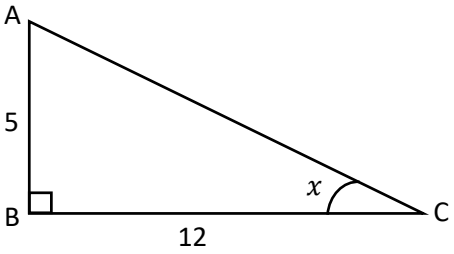
$24^2 + (\overline{NR})^2 = 30^2$

$(\overline{NR})^2 = 900 - 576 = 324$

$\overline{NR} = \sqrt{324}$

Therefore, $\overline{NR} = 18cm$……01 mark

(b)



$c^2 = a^2 + b^2$

$c^2 = 5^2 + 12^2 = 25 + 144 = 169$

$c = \pm\sqrt{169}\,, c = 13, c \geq 0$ ……01 mark

$sinx = \frac{opposite}{hypotenuse} = \frac{5}{13}$……$00\frac{1}{2}$ mark

$cosx = \frac{adjacent}{hypotenuse} = \frac{12}{13}$ …… ….$00\frac{1}{2}$ mark

$\left(\frac{sinx+cosx}{2}\right) = \left(\frac{5}{13} + \frac{12}{13}\right) \times \frac{1}{2} = \frac{17}{26}$……01 mark

10. (a) i. . $a^2 - b^2 = (a + b)(a - b)$

$8999^2 - 7001^2 = (8\,999 + 7\,001)(8\,999 - 7\,001)$……$00\frac{1}{2}$ mark

$= 16\,000 \times 1\,998$……$00\frac{1}{2}$ mark

$= 31\,968\,000$……$00\frac{1}{2}$ mark

ii. Product is 10 and factors are 5,2

$= x^2 + 5x + 2x + 10$…… …$00\frac{1}{2}$ mark

$= x(x + 5) + 2(x + 5)$……$00\frac{1}{2}$ mark

$= (x + 2)(x + 5)$…… $00\frac{1}{2}$ mark

(b) $x^2 - 11x + 30 = 0$

$x^2 - 11x + \left(\frac{-11}{2}\right)^2 = -30 + \frac{121}{4}$

$x^2 - 11x + \left(\frac{-11}{2}\right)^2 = \frac{-120+121}{4}$

$x^2 - 11x + \left(\frac{-11}{2}\right)^2 = \frac{1}{4}\left(x - \frac{11}{2}\right)^2$

$x^2 - 11x + \left(\frac{-11}{2}\right)^2 = \pm\frac{1}{4}$ ……01 mark

$\sqrt{\left(x - \frac{11}{2}\right)^2} = \pm\sqrt{\frac{1}{4}}$

$x - \frac{11}{2} = \pm\frac{1}{2}$……01 mark

$x = \frac{11}{2} \pm \frac{1}{2}$

$x = \frac{12}{2}\ or\ x = \frac{10}{2}$

$x = 6\ or\ x = 5$ ……01 mark.

SECTION B (40 MARKS)

11. Frequency distribution table

Class intervals	tally	Frequency
$40 - 49$		12
$50 - 59$		6
$60 - 69$		28
$70 - 79$		4
Total ……..01mark.		50……..01 mark

i. Mode ; modal class is $60 - 69$……01 mark

Mode $= l + \left(\frac{t_1}{t_1 + t_2}\right) i$

$= 59.5 + \left(\frac{28-6}{28-6+28-4}\right) 10$

$= 59.5 + \left(\frac{22}{22+24}\right) 10$

$= 59.5 + \frac{220}{46}$……01 mark

$= 59.5 + 4.7826 \ldots$

$= 64.28$ (2 decimal places)……..01 mark

Mean by assumed mean : $54\frac{13}{26} = 54.5$……01 mark

Class intervals	frequency	x	$d = x - A$	fd
$40 - 49$	12	44.5	-10	-120
$50 - 59$	6	54.5	0	0
$60 - 69$	28	64.5	10	280
$70 - 79$	4	74.5	20	80
Total	50	01 mark.	01 mark.	$\varepsilon fd = 240 \ldots \ldots 01\ mark$

$\overline{X} = A + \frac{\varepsilon fd}{N}$

$= 54.5 + \frac{240}{50}$…… ……..01 mark

$= 54.5 + 4.8 = 59.3$……01 mark

12. (a) $R = \{(x, y): y \geq |x|\ , and\ y \leq 3\}$……

Domain of R $= \{-3 \leq x \leq 3\}$……$00\frac{1}{2}$ mark.

Range of R $= \{0 \leq y \leq 3\}$……$00\frac{1}{2}$ mark

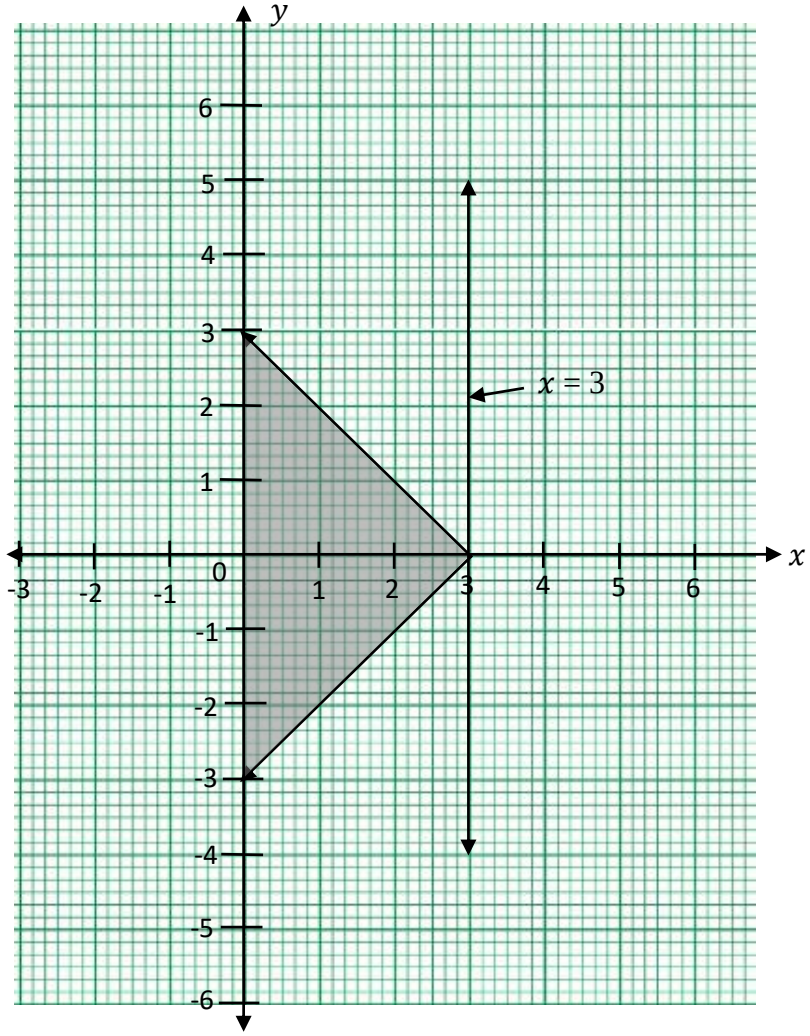
Domain R =Range of R^{-1}……$00\frac{1}{2}$ mark

Range of R = Domain of R^{-1}……$00\frac{1}{2}$ mark

Therefore Domain $R^{-1} = \{0 \leq x \leq 3\}$.....$00\frac{1}{2}$ mark

Range of $R^{-1} = \{-3 \leq y \leq 3\}$.....$00\frac{1}{2}$ mark

$R^{-1} = \{(x, y): y \leq |x|\ , and\ x \leq 3\}$.....01 mark



$1\frac{1}{2}$ Marks

(b) Domain of R $= \{All\ real\ numbers\}$.....01 mark

$y - 1 = 2x^2$

$\frac{y-1}{2} = \frac{2x^2}{2}$

$x^2 = \frac{y-1}{2}$

$\sqrt{x^2} = \sqrt{\frac{y-1}{2}}$

$x = \sqrt{\frac{y-1}{2}}$.....01 mark

$y - 1 \geq 0 \times 2$

Therefore range of R $= \{y: y \geq 1\}$.....01 mark

13. (a) i. $1\ cm = 25\ 000cm$

$60cm = y$

$y = \frac{25\ 000\ cm \times 60\ cm}{1\ cm}$

$y = 15\ 000\ 000\ cm$ ……..01 mark

$1\ km = 100\ 000\ cm$ ……..01 mark

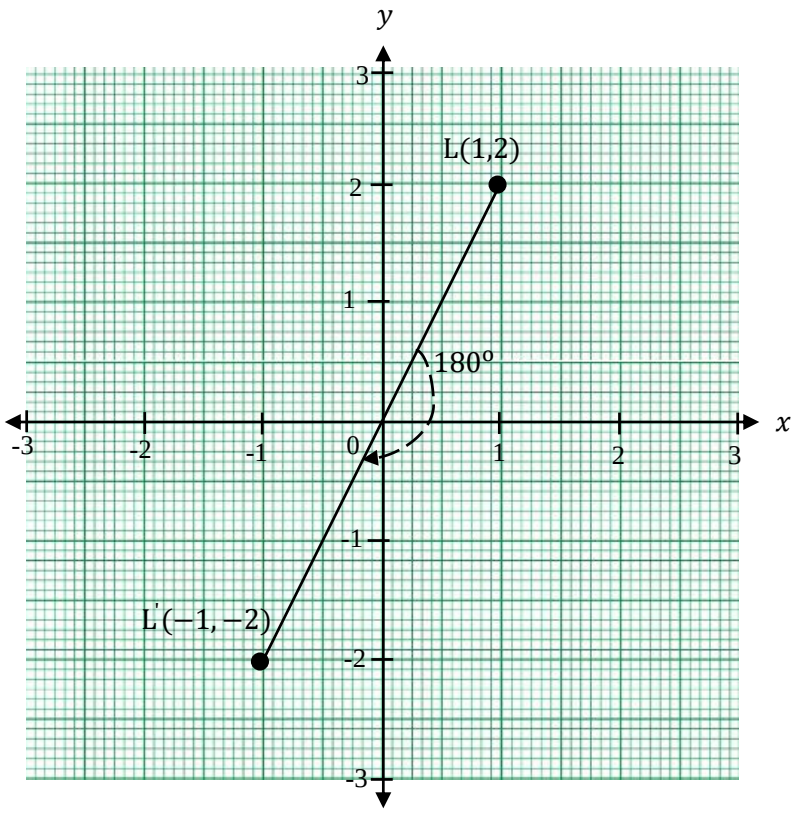
$x = 15\ 000\ 000\ cm$

$x = \frac{1\ km \times 15\ 000\ 000\ cm}{100\ 00\ cm}$

$= 15\ km$

Therefore the distance between point A and B on the ground is 15 km……01 mark

ii.



From the graph above

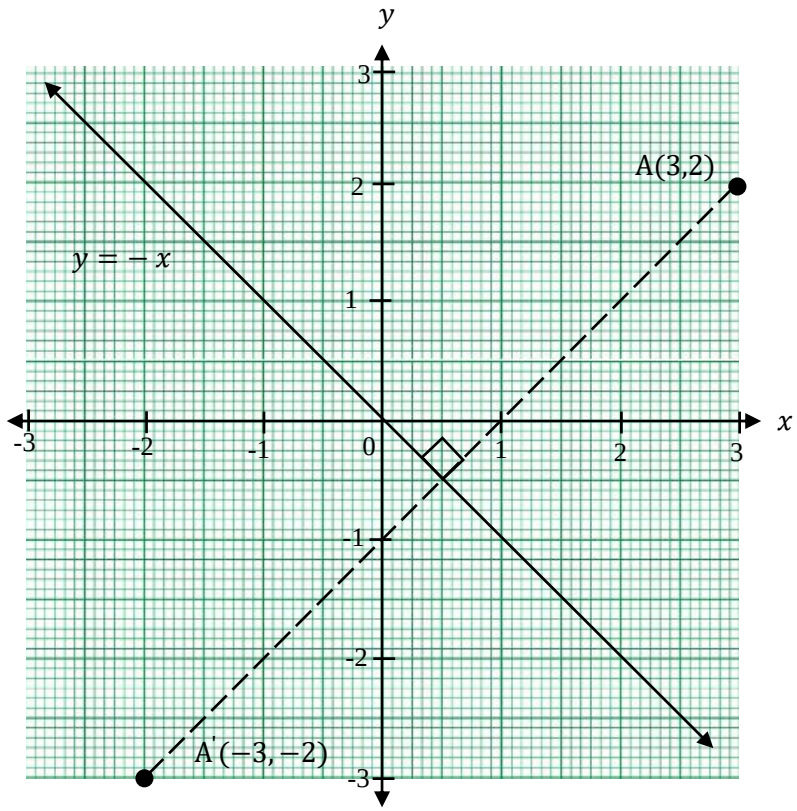
$R_{-180^\circ}(1,2) = (-1,-2)$

Generally, the rotation of a point through 180° clockwise about the origin is given by

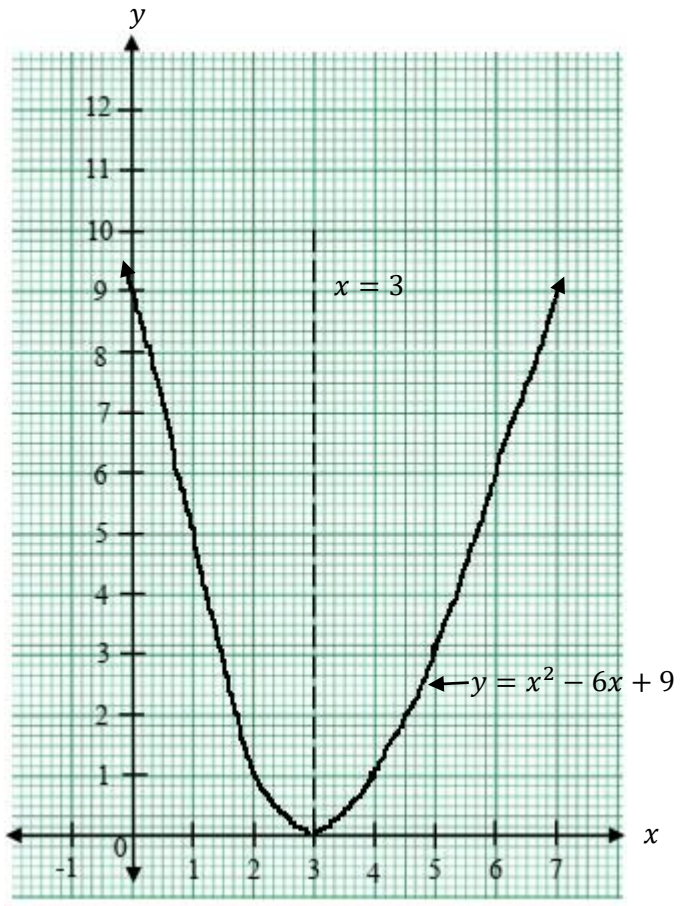
$R_{-180^\circ}(x,y) = (-x,-y)$

(b) The reflection of a point in the line $y = -x$ is shown in the graph below

Therefore, the reflection of the point A (2,3) in the line $y = -x\ is\ A'(-2,-3)$



14. (a) i.



....00 $\frac{1}{2}$ mark

ii

$x = \frac{-b}{2a} = \frac{6}{2} = 3$……$00\frac{1}{2}$ mark.

iii. $(x, y) = \frac{-b}{2a}, f\left(\frac{-b}{2a}\right)$

$3, 3^2 - 6 \times 3 + 9 = 0$

$(x, y) = (3,0)$…… $00\frac{1}{2}$ mark.

iv. Minimum point$y = f\left(\frac{-b}{2a}\right) or\ y = \frac{4ac-b^2}{4a} = 0$……$00\frac{1}{2}$ mark.

$y - intercept = c = 9\ x - intercept\ (y = 0)$

$x^2 - 6x + 9 = 0$

$x^2 - 3x - 3x + 9 = 0$

$x(x - 3) - 3(x - 3) = 0$

$(x - 3)(x - 3) = 0$

$x = 3$……01 mark

Domain of $f = \{All\ real\ numbers\}$……$00\frac{1}{2}$ mark.

Range of $f = \{y; y \geq 0\}$……$00\frac{1}{2}$ mark.

(b) $f(x) = \begin{cases} 4\ if\ x > 0 \\ 1\ if\ 1 < x \leq 3 \\ -3\ if\ x \leq\leq 1 \end{cases}$……03 marks

Domain of $f = \{All\ real\ numbers\}$……01 mark

Range of $f = \{4, 1, -3\}$……01 mark